Asian Journal of Environment & Ecology

Volume 24, Issue 11, Page 25-38, 2025; Article no.AJEE.146728
ISSN: 2456-690X

Trend Analysis of Global Energy Demand, Renewable Energy Production and Potential in India

R. Krishna [a], Mansi Vaid [a] and Manoj Sinha [a*]

[a] *Aryabhatta College, University of Delhi, Benito Juarez Road, New Delhi-110021, India.*

Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

Article Information

DOI: https://doi.org/10.9734/ajee/2025/v24i11815

Open Peer Review History:
This journal follows the Advanced Open Peer Review policy. Identity of the Reviewers, Editor(s) and additional Reviewers, peer review comments, different versions of the manuscript, comments of the editors, etc are available here: https://pr.sdiarticle5.com/review-history/146728

Review Article

Received: 17/08/2025
Published: 03/11/2025

ABSTRACT

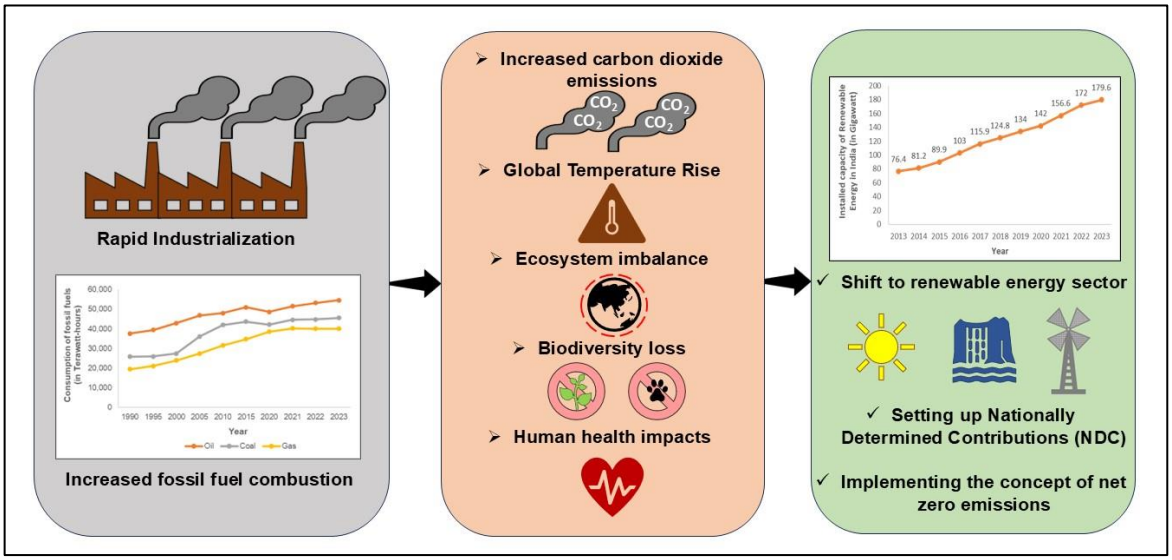
The continuously degrading state of the environment due to increasing fossil fuel consumption demands a quick overturn in the resource consumption patterns i.e., shift from non-renewable assets like fossil fuels to renewable assets. The present review focuses on this theme and evaluates the trends of fossil fuel consumption, carbon dioxide emission, temperature fluctuations, renewable energy transitions, and efforts to reduce climate change with a special emphasis on India. Data-driven trend analysis provides useful insights into the consumption patterns, environmental trends, and energy transition analysis, which can help in strategic planning and development of policies for sustainable growth. The Industrial Revolution has significantly contributed to climate change, primarily through the combustion of fossil fuels such as coal, oil, and natural gas for energy production. As a result, global temperatures have risen. If the current trajectory continues, average global temperatures may exceed pre-industrial levels by more than 1.5°C within the next two decades according to IPCC. To counter this, transitioning from fossil fuels to renewable energy is critical, particularly in the power sector, which is among the largest

**Corresponding author: E-mail: msinhadu@gmail.com, msinha@du.ac.in;*

Cite as: *R. Krishna, Mansi Vaid, and Manoj Sinha. 2025. "Trend Analysis of Global Energy Demand, Renewable Energy Production and Potential in India". Asian Journal of Environment & Ecology 24 (11):25–38. https://doi.org/10.9734/ajee/2025/v24i11815.*

contributors to CO_2 emissions worldwide. India is one of the leading emitters of CO_2 worldwide contributing 8.2% of the global emissions. To reduce its carbon footprint, India has developed around 179.6 GW of capacity from renewable energy sector. As per 2021 Nationally Determined Contributions (NDC), India has committed to fulfill nearly 50% of its electrical needs from non-fossil fuel-based energy resources by 2030. The Indian government is actively promoting renewable energy through various policies, incentives, and subsidies aimed at scaling up clean energy technologies. This review highlights these pressing concerns and emphasizes the importance of coordinated action by policymakers, industries, and civil society to mitigate climate change and secure a healthier environment for future generations.

GRAPHICAL ABSTRACT



Keywords: Global warming; climate change; fossil fuel; renewable energy resources; sustainability.

1. INTRODUCTION

Nearly 87% of the energy supplied to the world are fulfilled by fossil fuels like oil, coal, and natural gas (Energy Institute, 2025). These fuels have been the primary source of energy for industrial and transportation sectors for over a century due to their high energy density and relatively low cost of processing. However, there is a growing concern about the environmental and human health impacts of burning fossil fuels, such as air pollution, greenhouse gas emissions which contribute to global warming and climate change (Hassan et al., 2024).

Industries account for nearly two-fifth of the global energy consumption and around 30% of the energy is directly utilized in various industrial processes (Bashmakov et al., 2022). As global energy demand continues to rise, the finite capacity of fossil fuel reserves raises concerns about future availability and supply security. Depletion of easily accessible reserves can lead to higher extraction costs and geopolitical tensions over remaining reserves. Furthermore, combustion of fossil fuel emits CO_2, which severely affect the climate. The levels of CO_2 have increased by approximately 1.7 times from 1990 to 2022 (Energy Institute, 2024).

Climate change is a global issue which is defined as the long-term shifts in temperature and weather patterns of a region. It mainly occurs due to the emissions of greenhouse gases (GHG) from both natural processes (decomposition of organic waste) and anthropogenic activities. Major greenhouse gases are carbon dioxide, methane, and nitrous oxide. Some other GHG includes chloroflourocarbons (CFC), hydochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFC), bromofluorocarbons, perfluorocarbon (PFC), and sulfur hexafluoride (SF_6) which are emitted from industrial processes

or manufactured for human uses. While there are several greenhouse gases, CO_2 is particularly significant due to its abundance and long-lasting nature. It is solely responsible for approximately 66% of the warming effect on the climate (World Meteorological Organization, 2021).

According to NOAA (National Oceanic and Atmospheric Administration) and other climate science organizations, global land and ocean temperatures have been rising consistently. This trend has been observed over the past several decades, with each decade showing increasing average temperatures compared to the previous one. As per NOAA (2023) report, global land and ocean temperature in 2023 was 1.18^{0}C higher than the 20th century average of 13.9^{0}C. Such increasing temperature would likely exacerbate existing climate change impacts. This includes more intense heatwaves, alteration in precipitation patterns which can lead to floods or droughts, and altered ecosystems. Warmer temperatures can induce melting of polar ice caps and glaciers, which contributes to rise in sea levels. Many species are sensitive to changes in temperature. A rise in temperature could lead to shifts in habitats and potential disruptions to ecosystems, affecting biodiversity and the services they provide.

Net-zero is a global goal which aimed at mitigating climate change and its impacts. International agreements, like the Paris Agreement 2015, encourage countries to set and pursue net-zero targets to limit global warming to less than 2°C, and ideally 1.5°C, above the pre-industrial levels. Several countries pledge to achieve net-zero emissions by the year 2050 as part of their commitment to the Paris Agreement 2015 (IPCC, 2018). As a result, there is increasing interest and investment in the renewable energy sources like solar, hydroelectric, geothermal, and wind power, which provide sustainable replacements to fossil fuels. The installed capacity of renewable energy technologies has been expanding rapidly. In 2023, nearly 510 gigawatts of energy were added to the global renewable energy sector, which has been accounted as one of the fastest rates of growth in the past two decades. Particularly, wind and solar power have contributed a maximum share of nearly 30% in the global electricity production in the year 2023, and this share continues to grow. As per the latest report published by the International Energy Agency (IEA), global renewable energy capacity has been predicted to reach 7,300 gigawatts (GW) by 2028, reflecting a substantial expansion in this sector (International Energy Agency, 2023).

The continuously degrading state of the environment and diminishing resources demands a quick overturn in the resource consumption patterns i.e., shift from non-renewable assets like fossil fuels to renewable assets. The present review focuses on this theme and evaluates the trends of fossil fuel consumption, carbon dioxide emission, temperature fluctuations, renewable energy transitions, and efforts to reduce climate change with a special emphasis on India.

2. THE CONSUMPTION TREND FOR FOSSIL FUELS

The demand for fossil fuels is increasing constantly due to population growth and rapid industrialization. Historically, the primary sources of energy have been the fossil fuels, particularly coal, which is used for electricity generation and industrial processes. Further, in the present times, the largest amount of world's energy needs are being met by oil. It is one of the major energy sources for the transportation sector. For decades, natural gas has lagged behind oil and coal as a source of energy but has undergone a significant transformation in recent years as a replacement of coal, emerging as a prominent player in the global energy landscape. According to Kober et al. (2020), the primary energy consumption trends have more than doubled in the past forty years from 270.5 to 580 EJ globally. It was further estimated that nearly 85% of this energy was derived from fossil fuels. In India, per-capita energy consumption has shown a steady decadal rise, increasing from 1,773 kWh/person in 1980 to 2,660 kWh in 1990 (an addition of ~900 kWh). It further rose to 3,521 kWh in 2000 (~860 kWh increase), 5,025 kWh in 2010 (~1,500 kWh increase), 6,268 kWh in 2020 (~1,240 kWh increase), and has now reached about 7,813 kWh/person in 2024 (~1,550 kWh increase) (Our World in Data, 2025).

As per the Energy Institute (2024), the global consumption of oil, coal, and gas has increased in between 1950-2019 while it has decreased in the year 2020 due to the Lockdown measures, travel restrictions, and economic slowdowns caused by the COVID pandemic. As countries gradually eased lockdown measures and economic activities resumed, consumption of fossil fuels began to recover and continued to increase in subsequent years (Fig. 1).

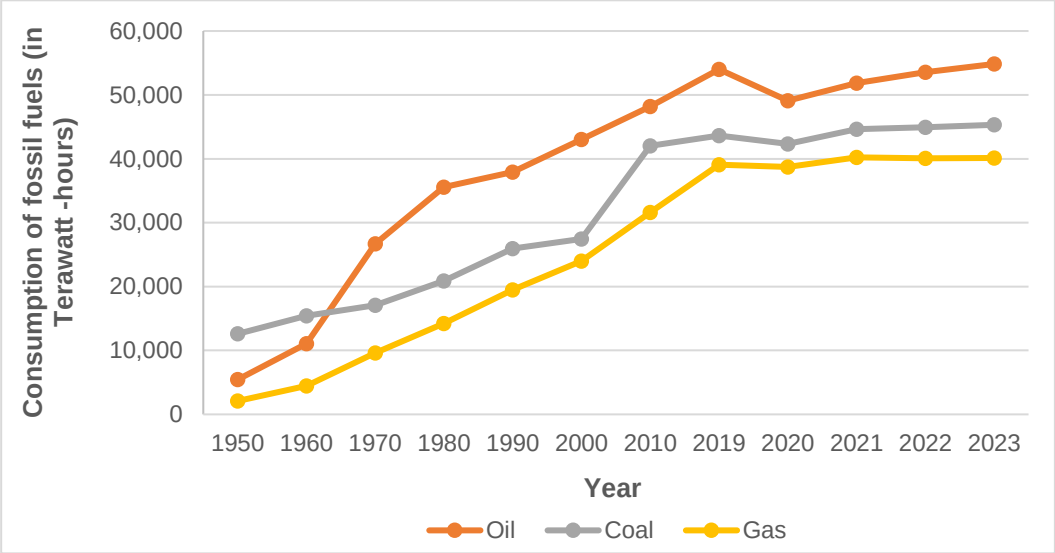


Fig. 1. Trends of global fossil fuel consumption. Data represents annual consumption of oil, gas, and coal in Terawatt-hours. All the data has been collected from Energy Institute (2024) and Our World in Data (2024)

The increasing energy demands are creating a severe pressure on the environmental resources and degrading the environmental quality. Specifically, in developing nations like India, which is one of the highest consumers of energy in the world (Aydin et al., 2016). Nearly half of the installed power generation capacity in India is derived from coal and around 50.52% of the energy demands is met by thermal power as per the Ministry of Power statistics. Owing to this scenario, the nation has earned the status of fourth highest polluting country in the world (PIB, 2025; Kumar and Majid, 2020). Over the years, this trend of fossil fuel consumption has seen a significant rise with 59% consumption in the year 1990, 67% in 2000, and 74% in 2010, post-which the values stabilized in their range showing slight variations (World Bank Group, 2025). As per the data records of Ministry of Coal (India), a large quantity of coal is imported in the country accounting to nearly 264.53 million tonnes in the year 2023-24. An estimated 243 million metric tonnes of crude oil and 27.7 million metric tonnes of liquified natural gas has been imported in the country in the financial year 2024-25 as per the official records of Petroleum Planning and Analysis Cell (Ministry of Petroleum & Natural Gas, Government of India). Increased consumption of fossil fuel releases large quantities of CO_2 in the atmosphere which is one of the leading causes for global warming.

3. VARIATIONS IN THE CO_2 EMISSIONS

Global CO_2 emissions have increased dramatically since the industrial revolution began in the late 18th century. In 2020, CO_2 emissions experienced a notable decline of approximately 5% due to the COVID-19 pandemic, the associated restrictions on economic activities, and mobility imposed by governments worldwide. In 2021, CO_2 emissions indeed rebounded past pre-pandemic levels, experiencing a growth rate of more than 5%. Emissions continued to increase, rising by 0.8% in 2022 and a further by 1.3% in 2023, ultimately reaching a new all-time high of 37.8 gigatons (Gt) in 2023 (Our World in Data, 2024) (Fig. 2). Globally, India stands as one of the top leaders of CO_2 emitters in the world, contributing nearly 8% of the world's total emissions in 2023 as per the statistics availed from Our World in Data (2024). A significant majority of carbon dioxide is emitted from electricity and industrial sources in the country, while the rest of the emissions are contributed by construction, agriculture, and transportation sectors.

While global CO_2 emissions have continued to rise, the rate of growth has slowed compared to previous decades, particularly the rapid increases seen during the 1970s and 1980s. This slowdown in the rate of emissions growth in the recent years has occurred despite ongoing global economic growth, indicating a decoupling of economic prosperity from carbon emissions. One

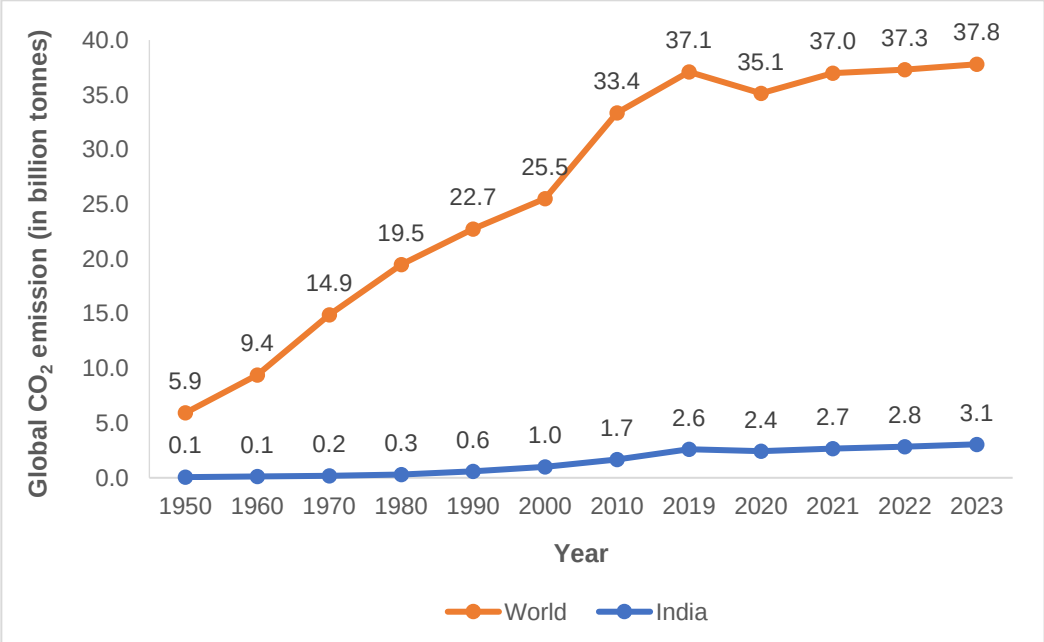


Fig. 2. Trends of global CO_2 emission from combustion and industrial processes. The data has been taken from Our World in Data (2024)

of the key factors contributing to this slowdown is indeed the increasing adoption of clean energy sources globally. As per the International Energy Agency statistics, many nations have increased their use of renewable energy, bioenergy, waste to energy, and nuclear energy, collectively considered as clean sources for the energy production. For instance, Brazil tops the list by adopting nearly 51.6% of clean energy resources, out of which 36.2% is contributed by bioenergy and waste, followed by France (50.3%), Indonesia (26.1%), Canada (24.6%), United Kingdom (23.2%), Germany (21.5%), Italy (21.3%), United States (20.3%), Korea (19.9%), Turkiye (18.1%), India (16.2%), and others (IEA, 2024). In India, efforts are being taken in terms of expansion of renewable energy sector and high efficiency projects like "Carbon Capture Utilization and Storage" to reduce the country's CO_2 emissions by 2050 and further achieve net-zero emission target by the year 2070 (Niti Aayog, 2022). Renewables and nuclear energy sources are generally categorized as low CO_2 emitting fuels. Global nuclear electricity generation exhibited a steady upward trajectory from 1970, attaining approximately 2,761 terawatt-hour (TWh) by 2006. Thereafter, the trend stabilized, with inter-annual fluctuations, and output remained broadly constant, reaching around 2,765 TWh in 2024 (Ember, 2025). It has been further reported that USA, France, Russia, China, and South Korea produce relatively higher amounts of the energy from nuclear sources. Such efforts have helped to diversify energy sources and reduce reliance on fossil fuels, leading to lower emissions growth rates.

Additionally, advancements in technology, declining costs of renewable energy, and supportive government policies have accelerated the transition to clean energy in many regions. This transition, coupled with improvements in energy efficiency across various sectors, has contributed to the slowdown in emissions growth. While this trend is promising, it's important to note that global CO_2 emissions continue to rise, albeit at a slower pace. Achieving meaningful emission reductions will require sustained efforts, increase international cooperation, and further scale up of clean energy investments and policies to accelerate the transition to a low-carbon economy.

4. GLOBAL TEMPERATURE RISE

The earth's global temperature has undergone fluctuations throughout its history. Earth's average temperature has risen by approximately 0.06^0C per decade since 1850. Since 1982, the rate of global temperature rise has accelerated significantly, with an average increase of about 0.20^0C per decade. This rate of warming is more than three times faster than the long-term

average, indicating a pronounced uptick in global temperature trends in recent decades (NOAA, 2023).

National Oceanic and Atmospheric Administration (NOAA) reported that the year 2023 was the warmest and 2016 was the second warmest year since global record keeping started in 1850 (NOAA, 2023). As per NOAA 2023 report, global land and ocean temperature in 2023 was 1.18^{0}C higher than the 20th century average. If the current global warming trend continues, it is projected that over the next 20 years, the global average temperature will exceed 1.5^{0}C above pre-industrial levels (World Meteorological Organization, 2024).

Such temperature anomalies highlight the urgency of addressing climate change and its associated impacts. Rising temperatures can lead to various consequences, including rising sea levels, frequent and severe weather phenomenon, imbalances in the ecosystems and biodiversity. It has been reported by NASA that the global sea level has increased by 4 inches as compared to the 1993 levels (NASA, 2025). Due to the changing climate, extreme weather phenomenon like tropical cyclones (Category 3-5) have intensified in the last four decades (Seneviratne et al., 2021). In developing nations like India, rising temperatures have reportedly increased water scarcity issues, induced extreme weather events, affected agricultural productivity, vector-borne diseases, and aggravated health issues like heat strokes, dehydration, cramps, and fatigue (Ravindra et al., 2024). Climate projections indicate that by the 2030s, the mean summer temperatures across India are likely to increase by approximately 1-2.5°C, while winter temperatures are expected to rise by over 1°C relative to the 1991-2019 climatological baseline, if the current anthropogenic trends continue (Center for Study of Science, Technology and Policy, 2022).

Efforts to mitigate climate change typically focus on switching to renewable energy sources reduction in greenhouse gas emissions, improvising the energy efficiency outputs, and implementing adaptation measures to cope with the changing climate. International collaboration and concerted action across sectors are crucial in addressing the challenges posed by global warming.

5. THE IMPORTANCE OF NET ZERO EMISSION

The concept of "net zero emissions" has become a focal point in international discussions on climate change, and it was a significant topic at the 26th United Nations Climate Change Conference of the Parties (COP26). COP26, held in Glasgow, Scotland, in November 2021, brought together world leaders, negotiators, activists, and experts to address the urgent need for collective action to combat climate change.

At COP26, many countries announced commitments to achieve net zero emissions by mid-century or earlier (International Energy Agency, 2023). With specific reference to India, the country has committed to achieve its net-zero emission target by 2070 (PIB Delhi, 2023). These commitments were encapsulated in updated or enhanced nationally determined contributions (NDCs) under the Paris Agreement. To achieve net zero emissions, societies need to significantly reduce their emissions of greenhouse gases, particularly carbon dioxide, methane, and nitrous oxide. This can be accomplished through various means, including transitioning to renewable energy sources, improving energy efficiency, adopting sustainable land use practices, and implementing carbon capture and storage technologies.

6. TRENDS IN GLOBAL RENEWABLE ENERGY PRODUCTION

Global renewable energy production has experienced tremendous growth in recent years (Xu et al., 2019). This growth can be attributed to various factors including technological advancements, oil crisis, declining costs of renewable energy technologies, supportive government policies, increasing environmental awareness, and the urgent need to mitigate climate change. The four major crisis of fuel oils observed between 1973-2008 in the world history acted as turning points for the countries forcing them to diversify their energy resources and transition towards a cleaner and more sustainable energy system (Kesicki, 2010; Archer, 1990). Fig. 3 shows how global renewable energy production has risen since 1990. Growth in renewable energy production during the period 1990-2000 was steady but relatively slow. With increasing energy demands and concerns about the environment, the production levels accelerated in the subsequent years.

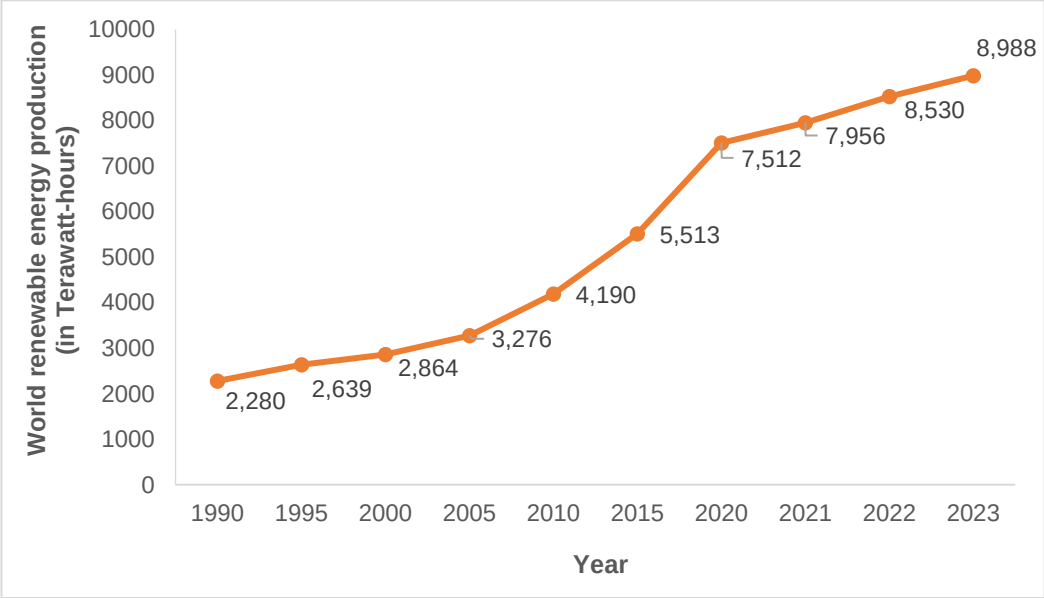


Fig. 3. Trends in world renewable energy production in Terawatt-hours. The data has been taken from Energy Institute (2024)

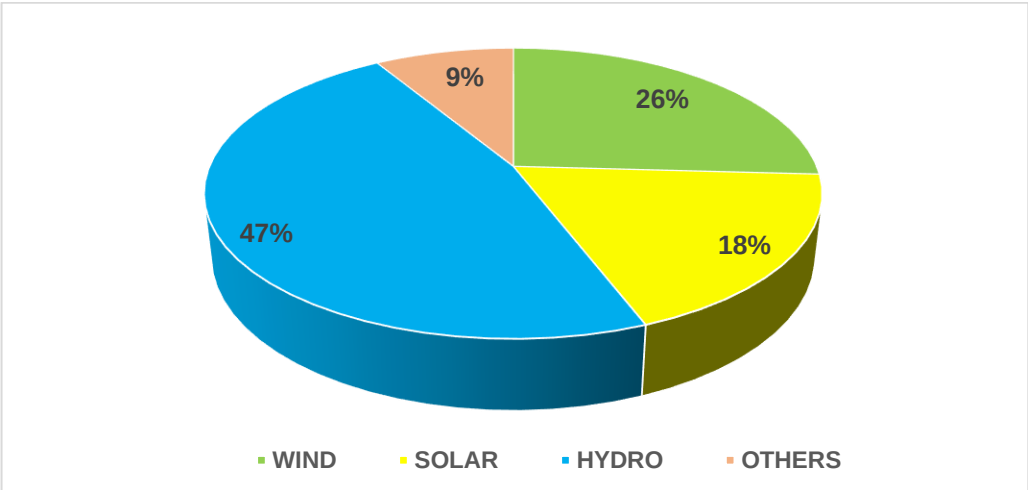


Fig. 4. Global contribution of different renewable energy resources in the year 2023. The data has been taken from Energy Institute (2024)

Renewable energy sources include solar, wind, hydroelectric, biomass, geothermal power, and others. However, hydropower is indeed a significant source of renewable energy worldwide. Many countries are increasingly focusing on solar and wind power as key components of their renewable energy strategies (Zabihi et al., 2025). In 2023, hydroelectricity shared approximately 47% of renewable energy production while wind and solar energy shared 26% and 18% respectively (Fig. 4).

As per International Energy Agency (2023), global annual renewable capacity addition reached to nearly 510 gigawatts in 2023 (Fig. 5). Among renewables, solar PV installations experienced an impressive twenty-sixfold increase from 2010 to 2022. By the end of 2022, global cumulative solar PV installed capacity reached 1047 GW, making it one of the fastest-growing renewable energy technologies. Though significant, current photovoltaic systems suffer from various challenges in their stability, efficiency, and energy density, which demand more research interventions in this direction

(Kumar et al., 2024b; Kumar et al., 2024c; Soni et al., 2024). Wind power installations also saw substantial growth, with a five-fold increase from 2010 to 2022. The global cumulative installed capacity for wind power (onshore) by the end of 2022, neared approximately 836 GW, indicating significant expansion in this sector. Global biofuels production (Bioethanol and Biodiesel) also increased 1.5-fold between 2013 to 2023 and reached to 2072 thousands of barrels of oil equivalent per day in 2023 (Energy Institute, 2024). Global hydrogen production capacity showed approximately 1.4-fold increment from 2013 to 2023 and reached to the level of 4,835 thousand tonnes in 2023 (Energy Institute, 2024).

Wind power installations also saw substantial growth, with a five-fold increase from 2010 to 2022. Hydropower constitutes the largest share of installed capacity for renewable power production. By 2022, global hydropower installed capacity (excluding pumped hydro) reached 1256 GW, accounting for 37% of the total renewable capacity. Other sources like geothermal, biomass, tidal, wave, and waste to energy sources, although started from a smaller base, experienced rapid growth over the past decade. The combined electricity production from these renewables reached 792.43 TWh in 2024 from 378.38 TWh levels in 2010 (Energy Institute - Statistical Review of World Energy, 2025).

The current progress in the renewable energy sector has been advocated by various initiatives taken by several countries across the world. For instance, the Germany government has introduced feed-in tariffs to promote and support renewable energy producers by guaranteeing them a fixed amount of income from the electricity generation (European Environment Agency, 2005-2012). Studies have predicted that the implementation of investment and production tax credits may increase solar energy generation by 10% in United States (EIA, 2022). Other schemes like renewable portfolio standards have been introduced in several countries like Australia, California, Korea, and Mexico. These standards help in achieving renewable energy targets to enhance their share in clean energy (Heeter et al., 2019).

Overall, these statistics underscore the significant progress made in transitioning towards a more sustainable and diversified energy mix. The growth of renewable energy technologies reflects increasing global efforts to reduce carbon emissions, mitigate climate change, and achieve energy security and sustainability goals (Demirbas, 2009).

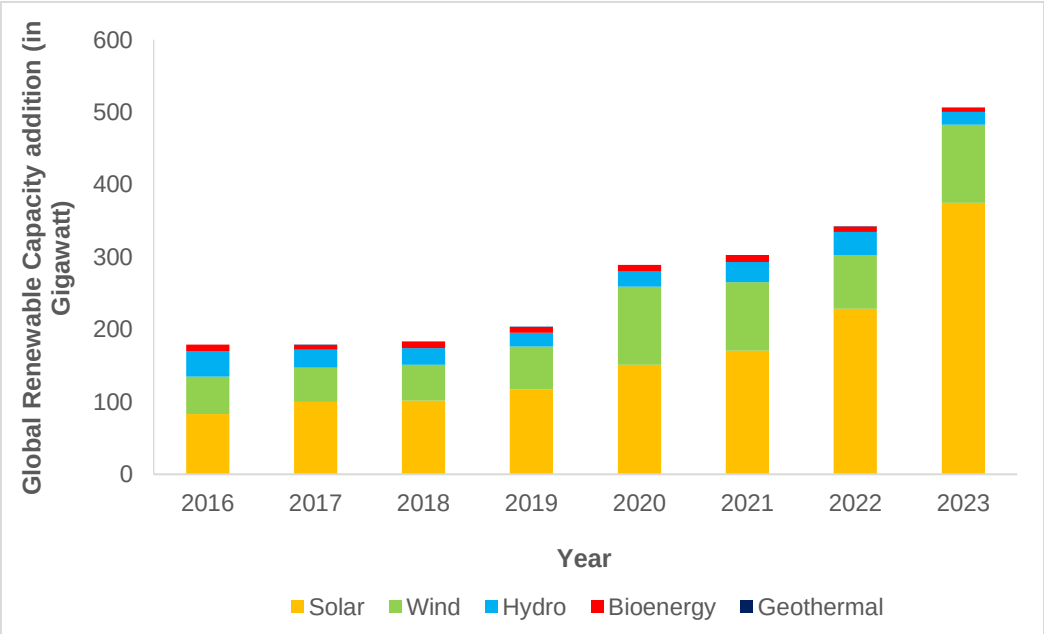


Fig. 5. Trends in the global renewable capacity addition for different sources of renewable energy. All data has been taken from International Energy Agency (2024)

7. INDIA'S NATIONALLY DETERMINED CONTRIBUTIONS (NDC) AND EFFORTS TO REDUCE CLIMATE CHANGE

India submitted its NDC on 2nd October 2015 to UNFCCC for the period of 2021-2030. As per the NDC, India has committed to reduce the emission intensity of its GDP by 33-35% by 2030 as compared to its 2005 level. The second important commitment was to achieve 40% of its installed electrical power production from non-fossil fuel-based resources by the year 2030. India also committed to increase its forest and tree cover to create an additional 2.5-3 billion tons of carbon sink by 2030. The country further updated its NDC at 26th UNFCCC conference (COP 26) held in Glasgow, United Kingdom in 2021. The country updated its commitment to reduce nearly 45% of its emissions intensity from GDP, in comparison to 2005 levels and also, achieve an installed capacity of nearly 50 percent from non-fossil fuel-based energy resources by the year 2030 (PIB Delhi, 2022).

The National Solar Mission was initiated in India in 2010 under the leadership of the then Prime Minister, with an initial objective of achieving 20 GW of solar power capacity by 2022. To strengthen climate change mitigation efforts and enhance electricity generation from renewable sources, the target was significantly revised in 2015, raising it to 100 GW (Press Information Bureau, 2021). India surpassed its original goal in 2018 and had reached an installed solar capacity of 63.30 GW by December 2022. As per the recent statistics, the current installed capacity of solar power is approximately 116 GW. Such progress is being achieved by various landmark initiatives being promoted across the country like PM Surya Ghar: Muft Bijli Yojana, Viability Gap Funding Scheme, and others (PIB Delhi, 2025). Apart from solar energy, the country is also harvesting electricity from other renewable resources such as wind, hydro etc. to reduce reliance on fossil fuels. As of December 2022, the estimated total wind power installed capacity of the country was 41.93 GW, and the total installed capacity of small/Micro/Mini hydro power was around 21,133.61 MW (Ministry of New and Renewable Energy, 2023a). The country is also working rapidly on the National Green Hydrogen Mission with an initial outlay plan of producing 5 MMT of green hydrogen by the year 2030 (Ministry of New and Renewable Energy, 2023b). India has also introduced policies for smart energy management initiatives in the areas of water and waste, conveyance, building, communal utilities, and regulations, which currently require better planning for effective implementation (Kumar et al., 2022). Five key priority areas namely robust grid and storage systems, power purchase agreement (PPAs), land-use optimization, enhanced access to finance, and domestic manufacturing have been strategized for successful implementation of the clean energy initiatives across the country (PIB Delhi, 2025).

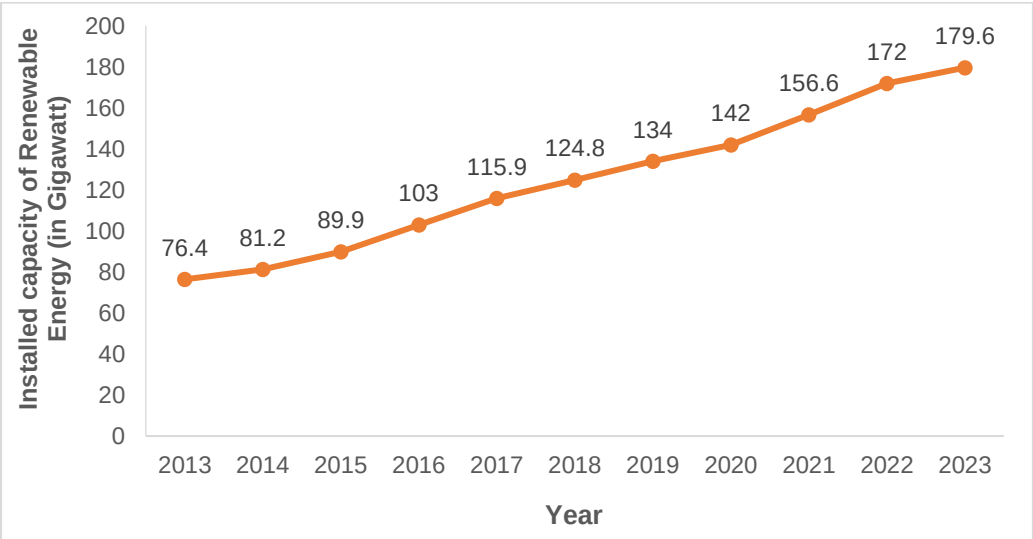


Fig. 6. Graphical representation of the installed capacity of renewable energy sources in India. All the data has been taken from Ministry of New and Renewable energy website

India exceeded its target of attaining 40% of installed electric power capacity from non-fossil fuel sources in November 2021, nearly nine years ahead of schedule (Ministry of New and Renewable Energy, 2022). By December 2022, the country's renewable energy capacity had grown to approximately 172 GW (Fig. 6), positioning India as the fourth largest country globally in this sector (Ministry of New and Renewable Energy, 2023a). As per the Climate change performance report, India ranked 8th in climate change performance index and it is the only G20 country in top 10performers (IPCC, 2022). Despite such advancements, certain challenges do exist in the implementation sector of renewable energy, like availability of a regulatory framework, state-wise renewable energy policies, penalty mechanisms in case of non-compliance, adequate research and development infrastructure, unavailability of human power, and others (Peng et al., 2022; Kumar and Majid, 2020).

8. CHALLENGES AND FUTURE RECOMMENDATIONS

Studying global energy trends can highlight areas where technological advancements are needed and helps policymakers to create effective regulations and incentives to promote sustainable energy practices and address energy challenges. Innovations in technology such as electric vehicles, smart grids, and sustainable energy initiatives like graphene-driven devices will help in altering the energy consumption patterns (Kumar et al., 2024a; Kumar et al., 2023). Promoting research in sustainable energy sector will ensure equitable access to renewable energy technologies, better understanding of the social implications of energy policies, and the economic impacts on various communities.

The future scope of trend analysis in global energy demand, renewable energy production, and India's potential in this sector involves a multidimensional approach. It encompasses understanding in the following areas:

- Energy demand patterns, technological advancements in renewable energy, the unique opportunities, and challenges in this sector.
- The role of international and regional cooperation.
- Understanding public attitudes and behaviors related to energy consumption and renewable energy adoption.
- Designing strategies to enhance public engagement, promote energy efficiency, and support the transition to sustainable energy practices.
- Regular assessment of energy consumption and renewable energy production helps track progress towards energy goals and targets, such as those set-in climate agreements or national energy plans.

9. CONCLUSIONS

In conclusion, while fossil fuels continue to meet a significant portion of global energy demand, the associated environmental, economic, and geopolitical challenges underscore the urgent need for transitioning to sustainable energy sources. Many governments have implemented policies and provided incentives to promote renewable energy. These include feed-in tariffs, tax credits, subsidies, and renewable portfolio standards that require a certain percentage of energy to come from renewables. In addition, research and development in renewable energy technologies have also led to improved efficiency and cost reductions. Innovations in solar panels, wind turbines, energy storage systems, and grid integration technologies have made renewable energy more competitive with fossil fuels. Despite these efforts, challenges remain, including intermittency issues with solar and wind power, energy storage solutions, grid modernization, and overcoming political and regulatory barriers. Continued commitment and collaboration at local, national, and international levels will be essential to accelerate the transition to a sustainable energy future powered by renewables. It is crucial to also allocate resources towards other renewable energy such as biofuels, hydropower, and geo-thermal energy to achieve a diversified and sustainable energy mix. The present review is an effort to synthesize a systematic data-driven knowledge perspective for this scenario across the world (with a special focus on India) so that the concerned policymakers and scientific community take effective actions to resolve this issue.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

REFERENCES

Archer, L., Barnes, P., Caffarra, C., Dargay, J., Horsnell, P., van der Linde, C., Skeet, I., & Al-Yousuf, A. (1990). *The first oil war—Implications of the Gulf crisis in the oil market* (Publisher's version). Oxford Institute for Energy Studies. Online at: https://ora.ox.ac.uk/objects/uuid:5b4a8a4e-bc45-48f3-ab42-bf8f29469be8/files/m3641c044c9ef3f6aab64358d9d2095b9; Date of access: 21st August, 2025.

Aydin, G., Jang, H., & Topal, E. (2016). Energy consumption modeling using artificial neural networks: The case of the world's highest consumers. *Energy Sources, Part B: Economics, Planning, and Policy, 11*(3), 212–219. https://doi.org/10.1080/15567249.2015.1075086; Date of access: 2nd March, 2025.

Bashmakov, I. A., Nilsson, L. J., Acquaye, A., Bataille, C., Cullen, J. M., de la Rue du Can, S., Fischedick, M., Geng, Y., & Tanaka, K. (2022). In IPCC: *Climate change 2022: Mitigation of climate change*, Chap. Industry. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (P. R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, & J. Malley, Eds.). Cambridge University Press. Online at: https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Chapter11.pdf; Date of access: 24th August, 2025.

Center for Study of Science, Technology, and Policy. (2022). *Climate atlas of India: District-level analysis of historical and projected climate change scenarios.* Online at: https://cstep.in/publications-details.php?id=2282; Date of access: 20th March, 2025.

Demirbas, A. (2009). Global renewable energy projections. *Energy Sources, Part B*, 4(2), 212–224. https://www.tandfonline.com/doi/abs/10.1080/15567240701620499; Date of access: 23rd October, 2025.

EIA. (2022). *Extending investment tax credits through 2050 increases U.S. solar generation by 10%.* Online at: https://www.eia.gov/todayinenergy/detail.php?id=53219; Date of access: 22nd February, 2025.

Ember. (2025). *Energy Institute - Statistical Review of World Energy (2025) – with major processing by Our World in Data.* Online at: https://ourworldindata.org/nuclear-energy; Date of access: 22nd August, 2025.

Energy Institute - Statistical Review of World Energy – with major processing by Our World in Data. (2025). Online at: https://ourworldindata.org/renewable-energy; Date of access: 10th September, 2025.

Energy Institute. (2024). *Statistical review of world energy.* Online at: https://www.energyinst.org/statistical-review/resources-and-data-downloads; Date of access: 15th January, 2025.

Energy Institute. (2025). *Statistical review of world energy.* Online at: https://www.energyinst.org/statistical-review/resources-and-data-downloads; Date of access: 15th January, 2025.

European Environment Agency. (2005–2012). *Country profile – Germany.* Online at: https://www.eea.europa.eu/en/analysis/publications/energy-support-measures/germany-country-profile; Date of access: 16th February, 2025.

Hassan, Q., Viktor, P., Al-Musawi, T., Ali, B. M., Algburi, S., Alzoubi, H. M., Al-Jiboory, A. K., Sameen, A. Z., Salman, H. M., & Jaszczur, M. (2024). The renewable energy role in the global energy transformations. *Renewable Energy Focus, 48,* 1–16. https://doi.org/10.1016/j.ref.2024.100545; Date of access: 16th February, 2025.

Heeter, J., Speer, B., & Glick, M. B. (2019). *International best practices for implementing and designing renewable portfolio standard (RPS) policies.* Golden, CO: National Renewable Energy Laboratory. NREL/TP-6A20-72798. Online at: https://www.nrel.gov/docs/fy19osti72798.pdf; Date of access: 6th February, 2025.

IEA. (2024). *Total energy supply for G20 countries by energy source, 2022.* IEA, Paris. https://www.iea.org/data-and-statistics/charts/total-energy-supply-for-

g20-countries-by-energy-source-2022 (Accessed 21 August 2025)

Intergovernmental Panel on Climate Change (IPCC). (2018). *Global Warming of 1.5°C: An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty* (V. Masson-Delmotte, P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P. R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J. B. R. Matthews, Y. Chen, X. Zhou, M. I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, & T. Waterfield, Eds.). Cambridge University Press. https://www.ipcc.ch/site/assets/uploads/sites/2/2022/06/SR15_Full_Report_HR.pdf (Accessed 24 August 2025)

Intergovernmental Panel on Climate Change (IPCC). (2022). *Climate change: A threat to human wellbeing and health of the planet. Taking action now can secure our future.* https://www.ipcc.ch/2022/02/28/pr-wgii-ar6/ (Accessed 21 December 2024)

International Energy Agency. (2023). *Renewables 2023: Analysis and forecast to 2028.* https://iea.blob.core.windows.net/assets/96d66a8b-d502-476b-ba94-54ffda84cf72/Renewables_2023.pdf (Accessed 4 January 2025)

International Energy Agency. (2024). *Emissions factors 2024.* https://www.iea.org/data-and-statistics/data-product/emissions-factors-2024-available-september-2024 (Accessed 4 January 2025)

Kesicki, F. (2010). The third oil price surge – What's different this time? *Energy Policy, 38*(3), 1596–1606. https://doi.org/10.1016/j.enpol.2009.11.044 (Accessed 21 August 2025)

Kober, T., Schiffer, H.-W., Densing, M., & Panos, E. (2020). Global energy perspectives to 2060 – WEC's World Energy Scenarios 2019. *Energy Strategy Reviews*, 31, 1-19. https://doi.org/10.1016/j.esr.2020.100523 (Accessed 26 February 2025)

Kumar, J. C. R., & Majid, M. A. (2020). Renewable energy for sustainable development in India: Current status, future prospects, challenges, employment, and investment opportunities. *Energy, Sustainability and Society, 10*(2). https://doi.org/10.1186/s13705-019-0232-1 (Accessed 12 December 2024)

Kumar, S., Banshwar, A., Sharma, N., Sharma, B. B., & Pathak, M. (2022, February). Strategies on smart energy management policies in India. In *Proceedings of the International Conference on Computational Intelligence and Sustainable Technologies: ICoCIST 2021* (pp. 637–645). Springer Nature Singapore. https://doi.org/10.1007/978-981-16-6893-7_56 (Accessed 24 October 2025)

Kumar, S., Giri, J., Sharma, S. S., Gunaga, S. R., Sathish, T., Hasnain, S. M., & Zairov, R. (2024a). Predictive analysis for removing obstacles in electric mobility: Revolution into EV adoption. *Transportation Engineering, 18,* 100277. https://doi.org/10.1016/j.treng.2024.100277 (Accessed 24 October 2025)

Kumar, S., Sharma, S. S., Giri, J., Makki, E., Sathish, T., & Panchal, H. (2024b). Perovskite materials with improved stability and environmental friendliness for photovoltaics. *Frontiers in Mechanical Engineering, 10,* 1357087. https://doi.org/10.3389/fmech.2024.1357087 (Accessed 24 October 2025)

Kumar, S., Supriya, J., Sharma, S. S., Paliwal, H., Manikanta, G., Giri, J., Hasnain, S. M. M., & Zairov, R. (2024c). Developments, challenges, and projections in solar battery charging in India. *Results in Engineering, 24,* 103248. https://doi.org/10.1016/j.rineng.2024.103248 (Accessed 24 October 2025)

Kumar, S., Yashaswini, H. K., Trupti, V., Rani, H. J., & Raghu, N. (2023). A review of the current research on graphene and its promising future. In *Low power designs in nanodevices and circuits for emerging applications* (pp. 105–120). https://doi.org/10.1201/9781003459231 (Accessed 24 October 2025)

Ministry of New and Renewable Energy. (2022). *Annual report 2021–22.* https://mnre.gov.in/annual-reports-2021-22/ (Accessed 21 December 2024)

Ministry of New and Renewable Energy. (2023a). *Annual report 2022–23.* https://mnre.gov.in/annual-reports-2022-23/ (Accessed 21 December 2024)

Ministry of New and Renewable Energy. (2023b). *National Green Hydrogen Mission.*

https://mnre.gov.in/en/national-green-hydrogen-mission/?utm (Accessed 21 August 2025)

NASA. (2025). *Sea level.* Retrieved August 21, 2025, from https://climate.nasa.gov/vital-signs/sea-level/?intent=121

Niti Aayog. (2022). *Carbon capture, utilization and storage (CCUS) policy framework and its deployment mechanism in India.* Retrieved August 24, 2025, from https://www.niti.gov.in/sites/default/files/2022-12/CCUS-Report.pdf

NOAA. (2023). *Global climate report 2023.* Retrieved December 21, 2024, from https://www.ncei.noaa.gov/access/monitoring/monthly-report/global/202313

Our World in Data. (2024). Retrieved January 15, 2025, from https://ourworldindata.org/fossil-fuels

Our World in Data. (2025). Retrieved August 22, 2025, from https://ourworldindata.org/grapher/per-capita-energy-use?time=latest&mapSelect=~IND

Peng, L., Zabihi, A., Azimian, M., Shirvani, H., & Shahnia, F. (2022). Developing a robust expansion planning approach for transmission networks and privately-owned renewable sources. *IEEE Access, 11,* 76046–76058. https://doi.org/10.1109/ACCESS.2022.3226695

PIB Delhi. (2022). *Cabinet approves India's updated nationally determined contribution to be communicated to the United Nations Framework Convention on Climate Change.* Retrieved August 22, 2025, from https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1847812&utm

PIB Delhi. (2023). *Ministry of Science & Technology. India is committed to achieve the Net Zero emissions target by 2070 as announced by PM Modi, says Dr. Jitendra Singh.* Retrieved April 11, 2025, from https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1961797

PIB Delhi. (2025). *Union Minister Pralhad Joshi highlights five pillars driving India's clean energy transformation.* Retrieved August 22, 2025, from https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147982&utm

PIB. (2025). *India's energy landscape: Powering growth with sustainable energy.* Retrieved August 22, 2025, from https://www.pib.gov.in/PressNoteDetails.aspx?NoteId=154717&ModuleId=3

Press Information Bureau. (2021). *Ministry of Information and Broadcasting. National Action Plan on Climate Change (NAPCC).* Retrieved January 22, 2025, from https://static.pib.gov.in/WriteReadData/specificdocs/documents/2021/dec/doc202112101.pdf

Ravindra, K., Bhardwaj, S., Ram, C., Goyal, A., Singh, V., Venkataraman, C., Bhan, S. C., Sokhi, R. S., & Mor, S. (2024). Temperature projections and heatwave attribution scenarios over India: A systematic review. *Heliyon, 10*(4). https://doi.org/10.1016/j.heliyon.2024.e26431

Seneviratne, S. I., Zhang, X., Adnan, M., Badi, W., Dereczynski, C., Di Luca, A., Ghosh, S., Iskandar, I., Kossin, J., Lewis, S., Otto, F., Pinto, I., Satoh, M., Vicente-Serrano, S. M., Wehner, M., & Zhou, B. (2021). Weather and climate extreme events in a changing climate. In V. Masson-Delmotte, P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, & B. Zhou (Eds.), *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (pp. 1513–1766). Cambridge University Press. https://doi.org/10.1017/9781009157896.013

Soni, P., Dave, V., Kumar, S., & Paliwal, H. (2024). A comparative study of AI-driven techno-economic analysis for grid-tied solar PV–fuel cell hybrid power systems. *The Scientific Temper, 15*(02), 2248–2257. https://doi.org/10.58414/SCIENTIFICTEMPER.2024.15.2.38

World Bank Group. (2025). *Fossil fuel energy consumption (% of total) – India.* Retrieved August 24, 2025, from https://data.worldbank.org/indicator/EG.USE.COMM.FO.ZS?locations=IN

World Meteorological Organization. (2021). *Greenhouse gas bulletin: Another year another record.* Retrieved March 24, 2025, from https://wmo.int/news/media-centre/greenhouse-gas-bulletin-another-year-another-record

World Meteorological Organization. (2024). *Climate change indicators reached record levels in 2023: WMO* [Press release]. Retrieved March 24, 2025, from https://wmo.int/news/media-centre/climate-change-indicators-reached-record-levels-2023-wmo

Xu, X., Wei, Z., Ji, Q., Wang, C., & Gao, G. (2019). Global renewable energy development: Influencing factors, trend predictions and countermeasures. *Resources Policy, 63,* 101470. https://www.sciencedirect.com/science/article/pii/S0301420719303174

Zabihi, A., Pidikiti, T., Varanasi, L. N. S., Kalnoor, G., Akkurt, I., & Krishna, V. B. M. (2025). Solar-powered pumping for alleviating energy poverty: Sustainable solutions for agriculture sector in Portugal. *IEEE Access, 13,* 105077–105090. https://doi.org/10.1109/ACCESS.2025.3579783

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of the publisher and/or the editor(s). This publisher and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

© Copyright (2025): Author(s). The licensee is the journal publisher. This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

Peer-review history:
The peer review history for this paper can be accessed here:
https://pr.sdiarticle5.com/review-history/146728